NÚRIA MORAGAS GARCIA, PhD

Bioinformatics Scientist | NGS | Multi-omics Data Science | Machine Learning

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PROFESSIONAL SUMMARY

PhD-trained Bioinformatician with 6+ years of experience in NGS data analysis, epigenomics, and multi-omics integration. Proven track record delivering reproducible, automated R/Python pipelines processing large-scale genomic and epigenetic datasets across multiple large-scale cohort studies. Skilled in translating complex biological data into actionable insights for precision medicine and cancer research. Experienced in machine learning-driven biomarker discovery and collaborating with cross-functional teams (clinicians, data scientists, experimental researchers) to support high-impact publications and data-driven decision making.

TECHNICAL SKILLS

Programming	Python (pandas, scikit-learn, matplotlib), R (tidyverse, Bioconductor), Bash, Git, Jupyter, RStudio, VS Code
Machine Learning	Random Forest, SVM, LASSO, PCA, Bayesian models, predictive modelling, feature selection, cross-validation
NGS & Pipelines	Nextflow, Snakemake, GATK, SAMtools, Bowtie2, Bismark, Kraken/Bracken, FastQC, MultiQC, Trim Galore
Data Types	WGS, RNA-seq, WGBS (DNA methylation), SNPs/GWAS, PRS, microbiome (shotgun metagenomics)
HPC & Cloud	SLURM job scheduling, Docker, Singularity/Apptainer, Linux/Unix systems, workflow automation
Visualization	ggplot2, Cytoscape, GraphPad Prism, Markdown, data storytelling for non-technical stakeholders
Statistics	GWAS, genotype imputation, population genetics, multi-omics integration, biomarker discovery, QC

PROFESSIONAL EXPERIENCE

IDIBELL Bioinformatics Data Scientist *L'Hospitalet de Llobregat, Spain*
Oncology Data Analytics Program (ODAP) Jul 2022 – Present

- Engineered automated R/Python bioinformatics pipelines for large-scale NGS data (WGS, RNA-seq, WGBS, microbiome), reducing manual processing time and improving reproducibility across multi-centre cohort studies.
- Analyzed genomic, transcriptomic, epigenomic, and metagenomic datasets (10,000+ samples) to identify genotype–phenotype associations, regulatory patterns, and cancer-related biomarkers.
- Applied ML models (Random Forest, LASSO, SVM, Bayesian) to prioritize candidate biomarkers from high-dimensional multi-omics datasets.
- Performed GWAS, genotype imputation, and large-scale genomic analyses using 1000 Genomes and TOPMed reference panels with rigorous QC and population genetics validation.
- Developed a Multi-Cancer Polygenic Risk Score (PRS) constellation model for cancer risk prediction.
- Collaborated with clinicians, experimental researchers, and data scientists to translate multi-omics results into clinical insights and data-driven reports.

IDIBAPS Bioinformatics & Biomedical Researcher *Barcelona, Spain*
Molecular and Translational Oncology Group Apr 2017 – Feb 2022

- Led computational investigation of transcriptional mechanisms in breast cancer progression (in situ → invasive), contributing to 3 first-author publications in high-impact journals.
- Built end-to-end RNA-seq analysis pipelines in R (FASTQ → QC → alignment → DE → functional enrichment), enabling reproducible analysis of tumour transcriptomic datasets.
- Integrated transcriptomic data with experimental results (RT-qPCR, immunofluorescence) to identify key regulatory pathways and candidate therapeutic targets.

Lawrence Berkeley National Laboratory Visiting Researcher *Berkeley, California, USA*
Laboratory of Mina Bissell – Biological Systems & Engineering Division Apr 2019 – Sep 2019

- Conducted hands-on research in 3D cell culture and genetic editing techniques within one of the world's leading cancer biology labs (Bissell Lab), recognised for pioneering work on tumour microenvironment.
- Applied experimental approaches to study tissue architecture and microenvironmental influences on cellular behaviour, complementing computational analyses of breast cancer progression.
- Gained experience in targeted genetic manipulation and advanced cell culture models, strengthening the translational bridge between experimental and bioinformatics work.
- Collaborated in a highly international, multidisciplinary team, developing scientific communication skills in English and exposure to US biomedical research culture.

GRIFOLS **Quality Control Supervisor** *Parets del Vallès, Spain*

Quality Control Department Apr 2015 – Apr 2017

- Led a team of 9 professionals, overseeing implementation of a new animal facility in compliance with ISO standards and regulatory requirements.
- Coordinated equipment validation, SOP documentation, and accreditation processes, improving operational efficiency by streamlining QC workflows.

EDUCATION

PhD in Biomedicine 2017 – 2021

Universitat de Barcelona (UB)

MSc in Bioinformatics and Biostatistics 2015 – 2023

UOC & Universitat de Barcelona (UB)

MSc in Translational Medicine 2014 – 2015

Universitat Autònoma de Barcelona (UAB)

BSc in Biology 2008 – 2013

Universitat de Barcelona (UB)

KEY PROJECTS & PORTFOLIO

- COLONOMICS (Multi-omics Integration). Integration of RNA-seq, miRNA and DNA methylation (WGBS) data from matched fresh normal, adjacent and tumour tissue samples to define molecular tissue signatures in colorectal cancer and identify epigenetically and transcriptionally distinct tissue states for biomarker discovery.
- Multi-Cancer PRS Constellation Model. Using genotyping data from GeneRisk/MCC-Spain and UK Biobank cohorts, built polygenic risk score (PRS) models across multiple cancer types and applied machine learning to construct a constellation model for population-level cancer risk stratification (MedRxiv 2024).
- GCATmb, Gut Microbiome of the Catalan Population. Extension of the GCAT cohort incorporating stool sample collection for shotgun metagenomics sequencing. Developed the full analysis pipeline and characterised the microbiome landscape of the Catalan population, identifying 3 main enterotype-like clusters associated with healthy and unhealthy lifestyle profiles (submitted to Microbiome).
- MARATO-COVID / Microbiome in Cancer Patients (collaboration with Vall d'Hebron Hospital). Shotgun metagenomics analysis comparing gut microbiota composition between cancer patients with and without COVID-19 infection. Investigated associations between microbiome dysbiosis, COVID-19 severity and mortality outcomes in oncological patients.
- Collaborative contributions. Provided bioinformatics support across multiple ODAP projects, including: (1) metabolomic and genetic validation of colorectal neoplasia biomarkers (Rius-Sansalvador et al., Cancer & Metabolism, submitted); (2) genome-wide interaction analysis of trihalomethane exposure and colorectal cancer risk (Moratalla-Navarro et al., Environment International 2026); (3) GWAS, Mendelian randomization and heavy metals exposure analysis; (4) genotyping-based PRS and diesel/air pollution exposure study; (5) TRIOS project, intergenerational epigenetics inheritance, analyzing CpG methylation patterns across parent-offspring trios to quantify the proportion of CpG sites explained by genetic heritability (meQTL-driven vs. environmentally determined). These contributions reflect a broad involvement across epidemiological, environmental and multi-omics research lines within the group.

SELECTED PUBLICATIONS

- Moragas N et al. **Multi-Cancer PRS Constellation Model for Cancer Risk Prediction.** MedRxiv, 2024.
- Moragas N et al. **The SEMA3F-NRP1/NRP2 axis is a key factor in invasive traits in in situ breast ductal carcinoma.** Breast Cancer Research, 2024.
- Moratalla-Navarro F*, Moragas N* et al. **Genome-Wide Interaction Analysis of trihalomethane exposure and colorectal cancer risk.** Environment International, 2026.
- Zubeldia-Plazaola A, Recalde-Percaz L*, Moragas N* et al. **Glucocorticoids promote transition of ductal carcinoma in situ to invasive ductal carcinoma.** Breast Cancer Research, 2018.

Full publication list available on request or via LinkedIn.

LANGUAGES

Catalan (Native) **Spanish** (Native) **English** (Advanced – C1)